

STAT C1000 — Exam 1 Practice Answer Key (Version P2)

Version P2 · Model answers (English) + brief notes (日本語 OK)

Question 1 — Variable classification (7 pts each)

Write-up format: (1) variable · (2) Qual/Quant · (3) subtype · (4) level

A) Variable: number of college courses enrolled in this semester

Quantitative · Discrete · Ratio

注: 科目数は数えられる整数。0科目は真のゼロ → discrete + ratio。

B) Variable: usual weekday wake-up time (military / 24-hour clock)

Quantitative · Continuous · Interval

注: 時刻 (clock time) 。差は意味あるが 0000 は「時間がない」ではない → interval。

C) Variable: distance from home to campus (miles)

Quantitative · Continuous · Ratio

注: 距離は測れる連続量。0マイルは真のゼロ、比も意味あり → continuous + ratio。

D) Variable: Likert agreement rating (1–5) about feeling prepared for exams

Qualitative · Ordinal · Ordinal

注: リッカート尺度。順序はあるが間隔が等しいとは限らない → qualitative + ordinal。

E) Variable: time to finish a typical STAT homework assignment

Quantitative · Continuous · Ratio

注: 所要時間 (duration) 。0は本当のゼロ → continuous + ratio。Bの「起床時刻」と混同しない。

Question 2 — CSULB coffee spending by class standing (24 pts)

A) Parameter: μ = the true mean amount of money CSULB students spend on coffee and cafe drinks per week. (Symbol 2 pts + description 2 pts)

注: 「true average」 = 母平均 μ 。by class standing なら各学年の μ でも可。 $\bar{x}$ と書かない。

B) All California State University, Long Beach students (the entire CSULB student body). (2 pts)

注: 母集団は CSULB の全学生。標本の250人ではない。

C) Variable: amount of money spent on coffee/cafe drinks per week. Quantitative, continuous, ratio level of measurement. (Description 4 pts + type/level 4 pts)

注: 金額は連続量で真のゼロあり → ratio。class standing は層分けの軸であり、関心変数ではない。

D) Sample: 250 students. Sampling method: stratified random sampling (strata = class standing; random sample from each freshman/sophomore/junior/senior group). (3 pts)

注: 学年で分けて各層から random = stratified。clusterなら「いくつかの学年を選んでその中を全部」になる。

E) Statistic: $\bar{x}$ (sample mean weekly coffee/cafe spending). Calculate: $\bar{x}$ = (sum of the 250 students' weekly spending amounts) / 250. (Symbol 4 pts + calculation 3 pts)

注: 統計量は標本から。学年別に出すなら各層内で $\bar{x}$ を計算。

Quick traps checklist

- military / 24-hour clock time → Interval (not ratio)
- Likert / agreement scales → Qualitative + Ordinal
- counts of courses → Discrete + Ratio
- distance / money / homework duration → Continuous + Ratio

- break into groups, sample from each → Stratified (not cluster)